Canonical Correlation Analysis

CETNEW Data — Question 14.4

Consumer Ethnocentric Tendencies and Attitudes Toward Importing Products

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1. Introduction and Study Context

This analysis examines the relationship between consumer ethnocentric tendencies (CET) and attitudes toward importing foreign products. The study involves **667 subjects** whose ethnocentric tendencies were measured using **seven indicators (X■–X■)**, and whose attitudinal responses toward importing were captured through **five measures (Y■–Y■)**. The data are provided as a covariance matrix in CETNEW.DAT.

Canonical Correlation Analysis (CCA) is the appropriate multivariate method here because it identifies the maximum correlations between linear combinations of the X set and linear combinations of the Y set. Rather than examining each X–Y pair in isolation, CCA finds the directions in both variable spaces that are most correlated, providing a structural model of how CET collectively relates to consumer attitudes.

Variable Sets

Set	Variables	Role	Count
X (Predictor)	X■, X■, X■, X■, X■, X■, X■	Ethnocentric tendency indicators (CET)	p = 7
Y (Criterion)	Y■, Y■, Y■, Y■, Y■	Attitudinal measures toward imports	q = 5

The number of canonical variate pairs is min(p, q) = min(7, 5) = **5 pairs**. Each pair (U■, V■) represents a dimension of association between the two variable sets, with the first pair capturing the strongest association.

2. Canonical Correlations

The five canonical correlations (r*) and their squared values are presented below. The squared canonical correlation (r^{*2}) indicates the proportion of shared variance between the k-th pair of canonical variates.

Pair	Canonical Corr (r*)	Squared (r^{*2})	Eigenvalue	Interpretation
r*1	0.759731	0.577191	0.577191	Very Strong
r*2	0.228468	0.052198	0.052198	Weak
r*3	0.186277	0.034699	0.034699	Weak
r*4	0.137212	0.018827	0.018827	Very Weak
r*5	0.084281	0.007103	0.007103	Negligible

The first canonical correlation r*■ = 0.7597 is by far the largest, with r*■² = 0.5772, meaning that the first pair of canonical variates shares **57.7% of variance**. This is a strong association by social science standards. The remaining four canonical correlations are all substantially smaller (below 0.23), indicating that the dominant relationship is captured almost entirely by the first canonical variate pair.

3. Significance Tests (Wilks' Lambda)

Wilks' Lambda (Λ) tests whether the remaining canonical correlations (from the i-th root onward) are jointly equal to zero. A small Λ indicates strong association. The chi-square approximation is: $\chi^2 = -[n - 1 - (p + q + 1)/2] \times \ln(\Lambda)$, giving a multiplier of 659.5 for this study.

Test (H■)	Wilks Λ	Chi-Square	df	p-value	Significance
Roots 1 to 5 = 0	0.376854	643.604	35	0.000000	*** (p < 0.001)
Roots 2 to 5 = 0	0.891312	75.882	24	0.000000	*** (p < 0.001)
Roots 3 to 5 = 0	0.940399	40.527	15	0.000377	*** (p < 0.001)
Roots 4 to 5 = 0	0.974203	17.236	8	0.027740	* (p < 0.05)
Roots 5 to 5 = 0	0.992897	4.701	3	0.195018	n.s. (p > 0.05)

Interpretation of significance tests:

• **Roots 1 to 5 (all roots):** $\Lambda = 0.3769$, $\chi^2(35) = 643.604$, $p < 0.001$ — The overall relationship between the CET indicators and the attitudinal measures is highly significant. There is strong evidence that CET collectively affects attitudes toward importing products.

• **Roots 2 to 5:** $\Lambda = 0.8913$, $\chi^2(24) = 75.882$, $p < 0.001$ — Even after removing the first canonical dimension, the remaining association is still statistically significant.

• **Roots 3 to 5:** $\Lambda = 0.9404$, $\chi^2(15) = 40.527$, $p = 0.0004$ — Still significant, though the association is much weaker.

• **Roots 4 to 5:** $\Lambda = 0.9742$, $\chi^2(8) = 17.236$, $p = 0.0277$ — Marginally significant at the 5% level.

• **Root 5 only:** $\Lambda = 0.9929$, $\chi^2(3) = 4.701$, $p = 0.1950$ — **Not significant.** The fifth canonical dimension does not contribute meaningful additional association beyond chance.

Conclusion: Four of the five canonical variate pairs are statistically significant ($p < 0.05$). However, practical significance is dominated by the first pair, which accounts for 57.7% of shared variance. Pairs 2–4 are statistically detectable largely because of the large sample size ($n = 667$) — their effect sizes ($r^{*2} < 0.055$) are small and should be interpreted cautiously.

4. Raw Canonical Coefficients

Raw canonical coefficients define the linear combinations of original variables that form the canonical variates. For the X set: U■ = a■X■ + a■X■ + … + a■X■. These coefficients are sensitive to the original scales and should be interpreted alongside the canonical structure coefficients.

4.1 X Set (Ethnocentric Tendency Indicators)

Variable	U1	U2	U3	U4	U5
X1	-0.0411	0.0325	-0.1988	0.0484	0.1457
X2	-0.0465	0.1187	0.5495	-0.0302	0.1681
X3	-0.1336	0.1485	0.0317	0.8685	-0.1331
X4	-0.0572	-0.4074	0.0259	0.0093	-0.2292
X5	-0.0764	0.1808	0.1326	-0.1758	-0.6225
X6	-0.2025	0.3502	-0.3523	-0.3256	0.1422
X7	-0.2429	-0.3790	-0.0665	-0.0729	0.3591

For the first canonical variate U■, all X coefficients are negative, indicating that higher ethnocentric tendencies on all seven indicators jointly move in the same direction along the first canonical dimension. X■ and X■ have the largest absolute coefficients (-0.2429 and -0.2025), suggesting they contribute most to U■.

4.2 Y Set (Attitudinal Measures)

Variable	V1	V2	V3	V4	V5
Y1	-0.3659	0.4178	0.2599	0.7512	-0.2442
Y2	-0.5190	0.3558	-0.6010	-0.3541	-0.3427
Y3	-0.2477	-0.5472	0.3793	-0.1696	-0.6831
Y4	-0.6923	-0.0773	0.3633	-0.2338	0.5727
Y5	-0.2368	-0.6273	-0.5434	0.4763	0.1683

For V■, all Y coefficients are also negative, consistent with the direction of U■. Y■ has the largest absolute coefficient (-0.6923), followed by Y■ (-0.5190), indicating these attitudinal measures are most strongly weighted in the first canonical dimension.

5. Canonical Structure Coefficients

Canonical structure coefficients (also called canonical loadings) are the correlations between the original variables and the canonical variates. They are more interpretable than raw coefficients because they are scale-free. Variables with |loading| > 0.30 are considered meaningful contributors to a canonical dimension.

5.1 X Variables with Canonical Variates U■–U■

Variable	U1	U2	U3	U4	U5
X1	-2.5975	-0.0071	-0.1330	0.2620	0.2502
X2	-2.3595	0.3046	1.3616	-0.1896	0.5063
X3	-2.8788	0.3733	-0.0642	1.7522	-0.1976

X4	-2.4188	-1.0958	0.2066	-0.0587	-0.5608
X5	-3.0045	0.2490	0.2093	-0.4648	-1.2547
X6	-3.5302	0.6954	-0.3914	-0.5085	0.1712
X7	-3.8297	-0.5975	-0.0352	-0.1462	0.3104

All seven X variables load strongly and in the same direction on U■, confirming that the first canonical variate captures a general ethnocentric tendency factor. X■ (loading = -3.8297) and X■ (-3.5302) are the strongest contributors. This means that consumers high on all seven CET indicators form a coherent attitudinal profile captured by U■.

5.2 Y Variables with Canonical Variates V■–V■

Variable	V1	V2	V3	V4	V5
Y1	-2.4329	0.1441	0.0131	1.1336	-0.5846
Y2	-2.9874	0.1442	-0.8678	-0.1504	-0.5024
Y3	-1.9850	-0.9738	0.4758	-0.0522	-1.2231
Y4	-3.3428	-0.5361	-0.0488	0.0580	0.3713
Y5	-2.4550	-1.1425	-0.9223	1.1352	0.2621

All five Y variables load strongly on V■. Y■ and Y■ show the highest loadings, indicating that these two attitudinal measures are most strongly associated with the dominant CET dimension. V■ represents a broad negative attitude toward imported products — consumers high on CET (high U■) are most negative toward imports (high V■ in the negative direction as defined by the sign of the coefficients).

6. Structural Model and Overall Interpretation

6.1 Structural Model Diagram

The structural model proposed by the study is:

CET Indicators (X■–X■)	→	Attitudinal Measures (Y■–Y■)
Ethnocentric tendency		Attitudes toward importing products
General CET factor (U■)	**r* = 0.7597**	**General attitude factor (V■)**

6.2 Summary of Findings

1. Strong primary association:

The first canonical correlation $r^*_{■} = 0.7597$ ($r^{*2}_{■} = 0.5772$) is statistically significant ($p < 0.001$) and practically meaningful. This means that 57.7% of the variance in the dominant attitude canonical variate is explained by the dominant CET canonical variate. CET strongly affects attitudes toward importing products.

2. Four significant dimensions:

Four of the five canonical variate pairs are statistically significant ($p < 0.05$). However, only the first pair has a practically meaningful effect size. Pairs 2–4 have r^{*2} values below 0.055, which are small effects detectable only due to the large sample size ($n = 667$).

3. General CET factor dominates:

All seven X indicators load in the same direction on U■, forming a general ethnocentric tendency construct. Similarly, all five Y indicators load consistently on V■. This confirms that CET operates as a unified psychological disposition that uniformly affects consumer attitudes toward imports.

4. Fifth canonical dimension not significant:

Root 5 ($r^*_{■} = 0.0843$) is not significant ($p = 0.1950$). The fifth canonical variate pair should be dropped from interpretation as it represents noise rather than a real dimension of association.

5. Practical conclusion:

Consumers with stronger ethnocentric tendencies (higher X■–X■ scores) hold more negative attitudes toward imported products (Y■–Y■). The structural model CET → Attitudes is confirmed and statistically robust with $n = 667$.

7. SAS and Python Implementation Notes

Two code files accompany this report:

cetnew_cca.sas — Uses PROC IML to manually construct the covariance matrix and compute canonical correlations via eigenvalue decomposition, followed by PROC CANCORR with a TYPE=COV input dataset. PROC CANCORR provides standardized coefficients, canonical structure, and redundancy indices automatically.

cetnew_cca.py — Uses NumPy and SciPy for all matrix operations. The covariance matrix is constructed from the lower triangular CETNEW.DAT values, submatrices S_XX, S_YY, S_XY are extracted, and canonical correlations are obtained as the square roots of the eigenvalues of S_XX^{-1/2} S_XY S_YY^{-1} S_YX S_XX^{-1/2}. Wilks' Lambda and chi-square tests are computed using the Bartlett approximation with multiplier $t = n - 1 - (p+q+1)/2 = 659.5$.